```
import kagglehub
aliiihussain_amazon_sales_dataset_path = kagglehub.dataset_download('aliiihussain/amazon-sales-dataset')

print('Data source import complete.')
```

```
import numpy as np
import pandas as pd
 this (by clicking run or pressing Shift+Enter) will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
/kaggle/input/datasets/aliiihussain/amazon-sales-dataset/amazon_sales_dataset.csv
```

Mathematical transformations

```
df=pd.read_csv('/kaggle/input/datasets/aliiihussain/amazon-sales-dataset/amazon_sales_dataset.csv')
```

```
df.head()
```

	order_id	order_date	product_id	product_category	price	discount_percent	quantity_sold	customer_region	payment_method
0	1	2022-04-13	2637	Books	128.75	10	4	North America	UPI
1	2	2023-03-12	2300	Fashion	302.60	20	5	Asia	Credit Card
2	3	2022-09-28	3670	Sports	495.80	20	2	Europe	UPI
3	4	2022-04-17	2522	Books	371.95	15	4	Middle East	UPI
4	5	2022-03-13	1717	Beauty	201.68	0	4	Middle East	UPI

```
df.info()
```

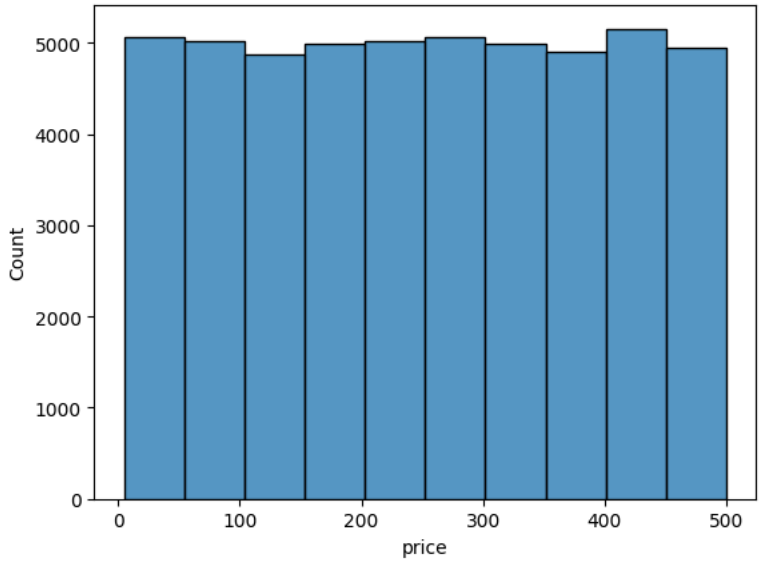
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_id          50000 non-null  int64
 1   order_date        50000 non-null  object
 2   product_id        50000 non-null  int64
 3   product_category  50000 non-null  object
 4   price             50000 non-null  float64
 5   discount_percent  50000 non-null  int64
 6   quantity_sold     50000 non-null  int64
 7   customer_region   50000 non-null  object
 8   payment_method    50000 non-null  object
 9   rating            50000 non-null  float64
 10  review_count      50000 non-null  int64
 11  discounted_price  50000 non-null  float64
 12  total_revenue     50000 non-null  float64
dtypes: float64(4), int64(5), object(4)
memory usage: 5.0+ MB
```

NORMAL DISTRIBUTION

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

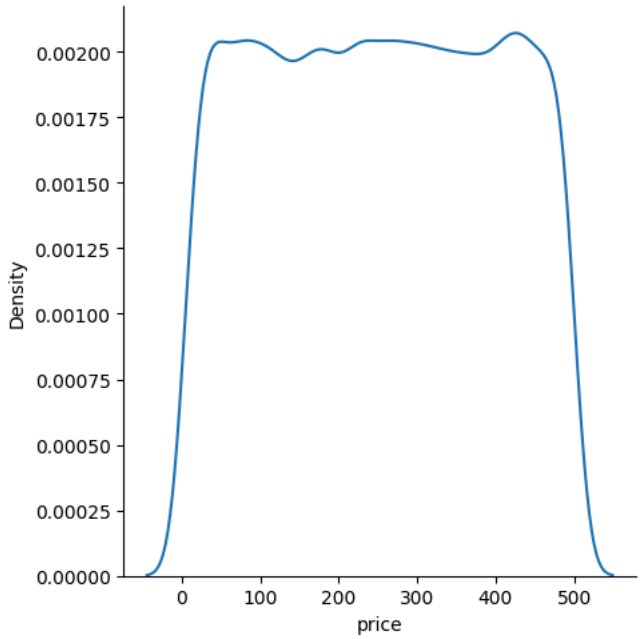
```
sns.histplot(df['price'],bins=10)
```

```
<Axes: xlabel='price', ylabel='Count'>
```



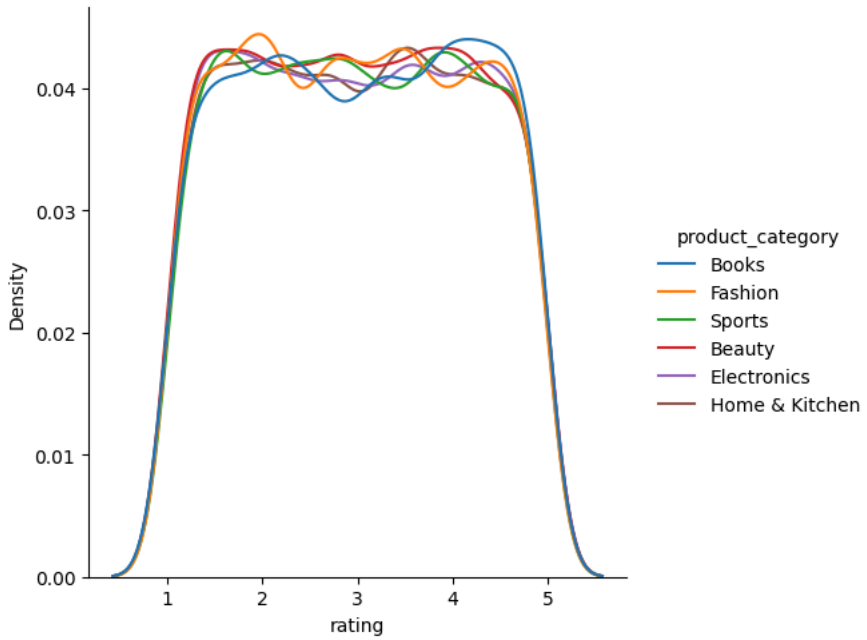

```
sns.displot(df['price'],kind='kde')
```

```
<seaborn.axisgrid.FacetGrid at 0x7de22e239040>
```



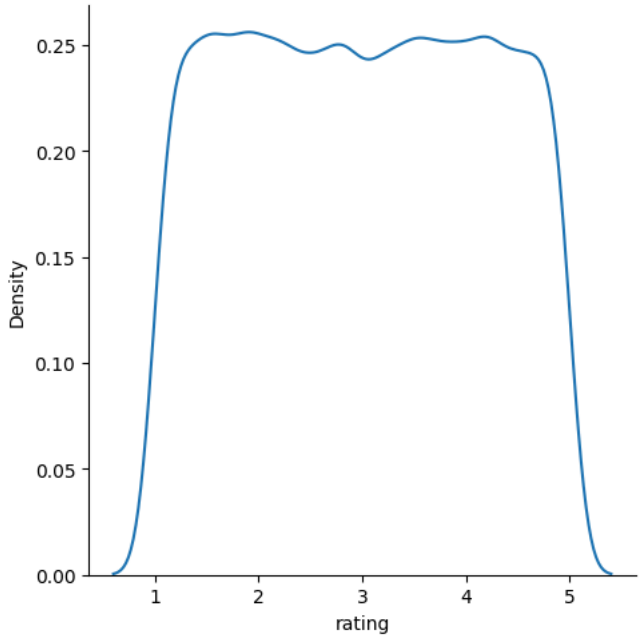

```
sns.displot(df,x='rating',hue='product_category',kind='kde')
```

```
<seaborn.axisgrid.FacetGrid at 0x78c0ebac6180>
```



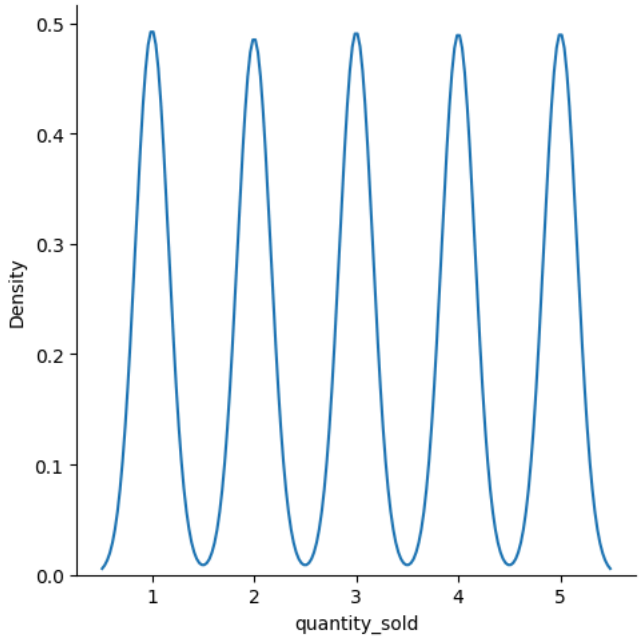

```
sns.displot(df,x='rating',kind='kde')
```

```
<seaborn.axisgrid.FacetGrid at 0x7de24a1aca40>
```



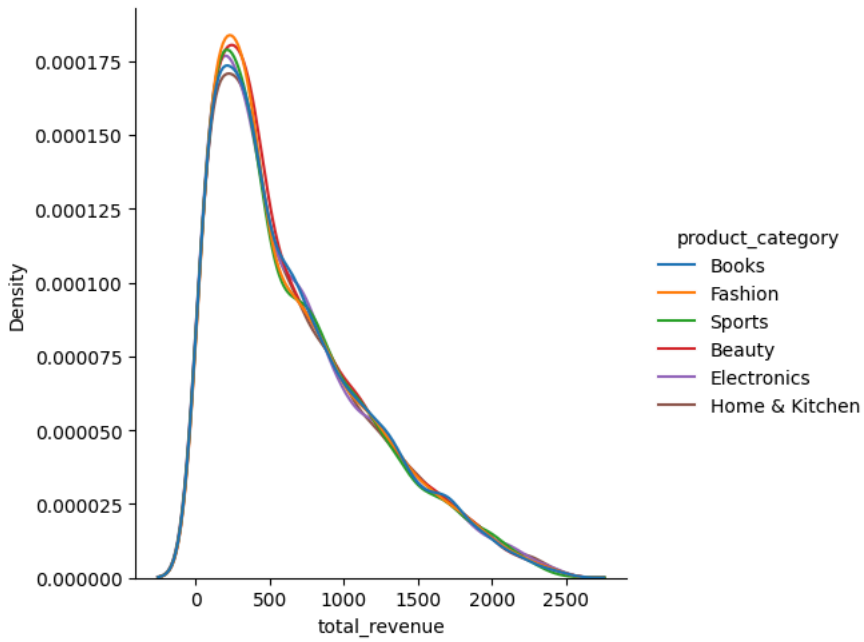

```
sns.displot(df,x='quantity_sold',kind="kde")
```

```
<seaborn.axisgrid.FacetGrid at 0x7de22e1431a0>
```




```
sns.displot(df,x='total_revenue',hue='product_category',kind="kde")
```

```
<seaborn.axisgrid.FacetGrid at 0x7de2328d24b0>
```



FUNCTION TRASFORMER

1} log transform

```
from sklearn.preprocessing import FunctionTransformer
import numpy as np
log_transform=FunctionTransformer(np.log1p)
transformed=log_transform.fit_transform(df['total_revenue'])
```

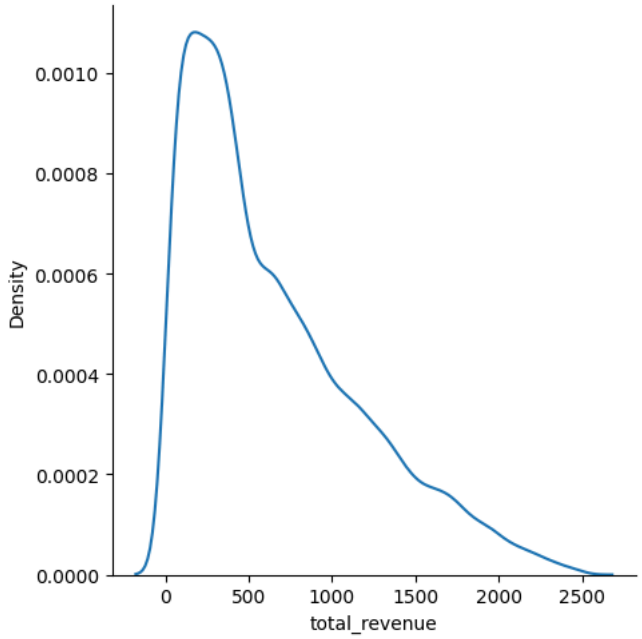
```
print(transformed)
```

```
0        6.141005
1        7.099532
2        6.677436
```

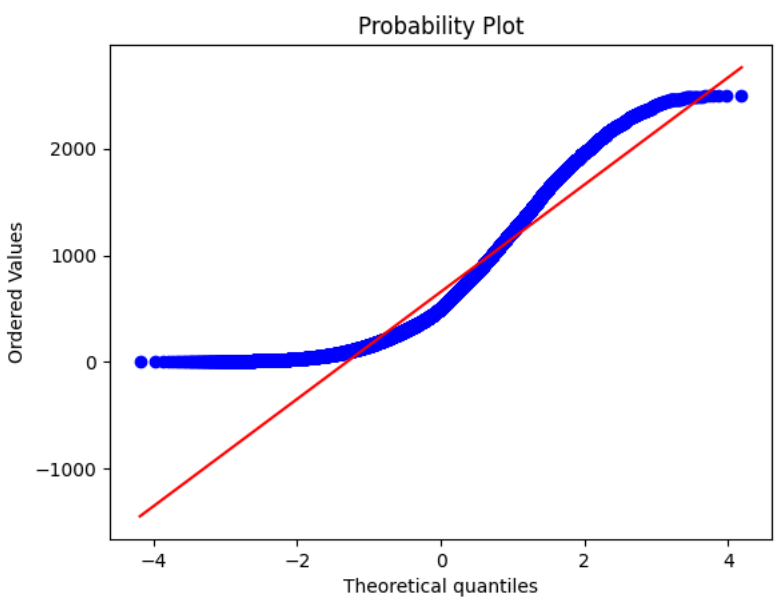
```
3        7.143333
4        6.694215
           ...
49995    4.912287
49996    7.189205
49997    6.894589
49998    5.680719
49999    5.184084
Name: total_revenue, Length: 50000, dtype: float64
```

```
sns.displot(df,x='total_revenue',kind='kde')
```

```
<seaborn.axisgrid.FacetGrid at 0x7de22e0bcbc0>
```




```
import matplotlib.pyplot as plt
import scipy.stats as stats
stats.probplot(df['total_revenue'],plot=plt)
plt.show()
```



POWER TRANSFORMER

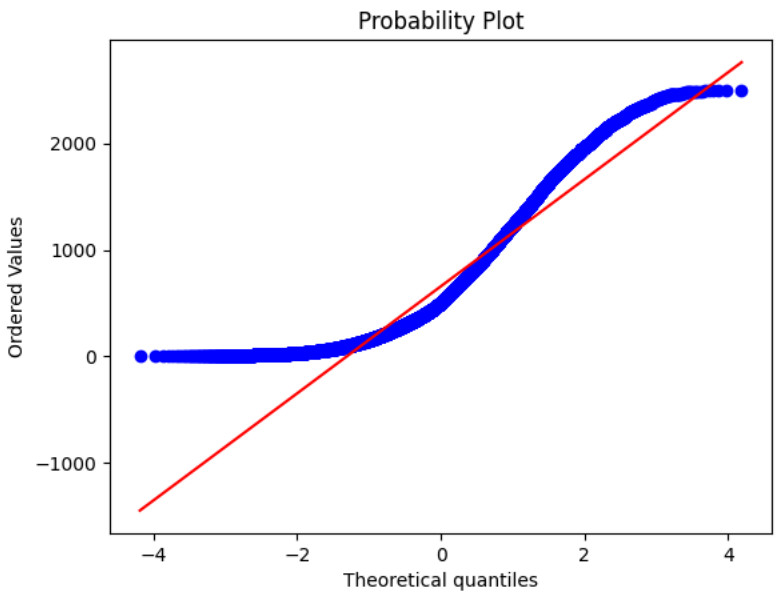
{1}BOX-FOX

```
from sklearn.preprocessing import PowerTransformer
pt_boxcox=PowerTransformer(method='box-cox')
data_transformed=pt_boxcox.fit_transform(df[['total_revenue']])
```

```
print(data_transformed)
```

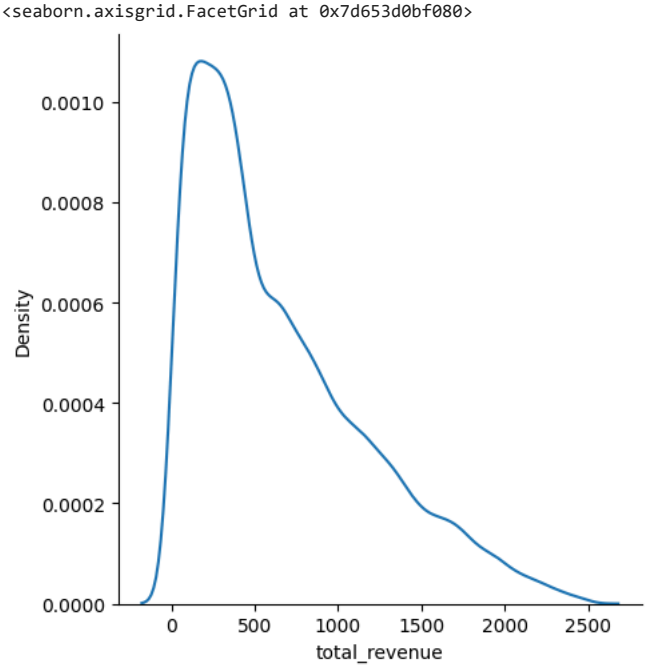
```
[[-0.1016889 ]
 [ 1.07321466]
 [ 0.5099516 ]
 ...
 [ 0.78994406]
 [-0.54705268]
 [-0.95789688]]
```

```
import matplotlib.pyplot as plt
import scipy.stats as stats
stats.probplot(df['total_revenue'],plot=plt)
plt.show()
```

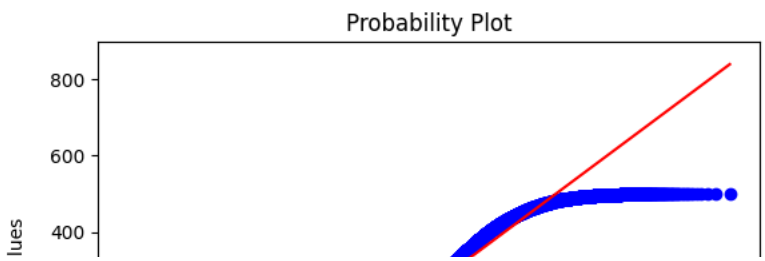



```
import seaborn as sns
sns.displot(df['total_revenue'],kind='kde')
```

```
<seaborn.axisgrid.FacetGrid at 0x7d653d0bf080>
```



{2}YEO-JOHNSON

```
from sklearn.preprocessing import PowerTransformer
pt=PowerTransformer(method='yeo-johnson')
data_transformed=pt.fit_transform(df[['price']])
```

```
import matplotlib.pyplot as plt
import scipy.stats as stats
stats.probplot(df['price'],plot=plt)
plt.show()
```